Logistic Regression

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Horseshoe Crab Mating Data

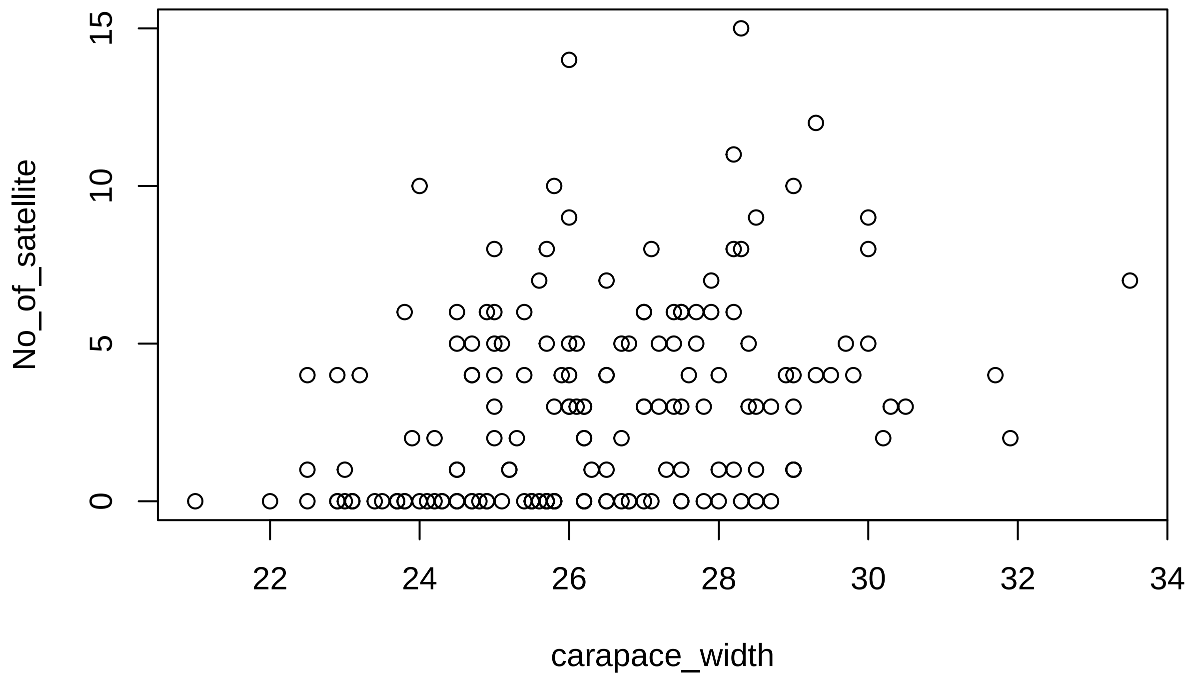
We illustrate the logistic regression using a study of female horseshoe crabs on an island in the Gulf of Mexico. During spawing season, the females migrate to a shore to breed, with a male attached to her posterioir spine, and she burrows into the sand and lays clusters of eggs. During spawing, other males crab may group around the pair and may also fertilize the eggs. These male crabs that cluster around the female crab are called satellites.

In this example, the response outcome of each of 173 female crabs is whether a female crab has a satellite or not. Explanatory variables are the female crab's color, spine condition, weight, and carapace width.
Color(1:light medium,2:medium,3:dark medium,4:dark). Spine condition(1:both good,2:one worn or broken,3:both worn or broken). Weight is in grams, carapace width is in cm, and y is whether a female crab has a satellite (1=yes, 0=no). For now we use width alone as the predictor. The sample mean width equals 26.3 and the standard deviation equals 2.1.

```
library(CatDataAnalysis)
data(table_4.3)
head(table_4.3)
```

```
##   color spine width satell weight y
## 1     3     3  28.3      8   3050 1
## 2     4     3  22.5      0   1550 0
## 3     2     1  26.0      9   2300 1
## 4     4     3  24.8      0   2100 0
## 5     4     3  26.0      4   2600 1
## 6     3     3  23.8      0   2100 0
```

```
carapace_width=table_4.3$width
No_of_satellite=table_4.3$satell
plot(carapace_width,No_of_satellite)
```




```
logistic.reg.model=glm(table_4.3$y~carapace_width,family = binomial(link = "logit"))
summary(logistic.reg.model)
```

```
##
## Call:
## glm(formula = table_4.3$y ~ carapace_width, family = binomial(link = "logit"))
##
## Coefficients:
##                Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -12.3508     2.6287  -4.698 2.62e-06 ***
## carapace_width   0.4972     0.1017   4.887 1.02e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 225.76  on 172  degrees of freedom
## Residual deviance: 194.45  on 171  degrees of freedom
## AIC: 198.45
##
## Number of Fisher Scoring iterations: 4
```

Binary response and one predictor

For a binary response variable Y and explanatory variable X, let $\pi(x) = P(Y = 1|X = x) = 1 - P(Y = 0|X = x)$. The logistic regression model is

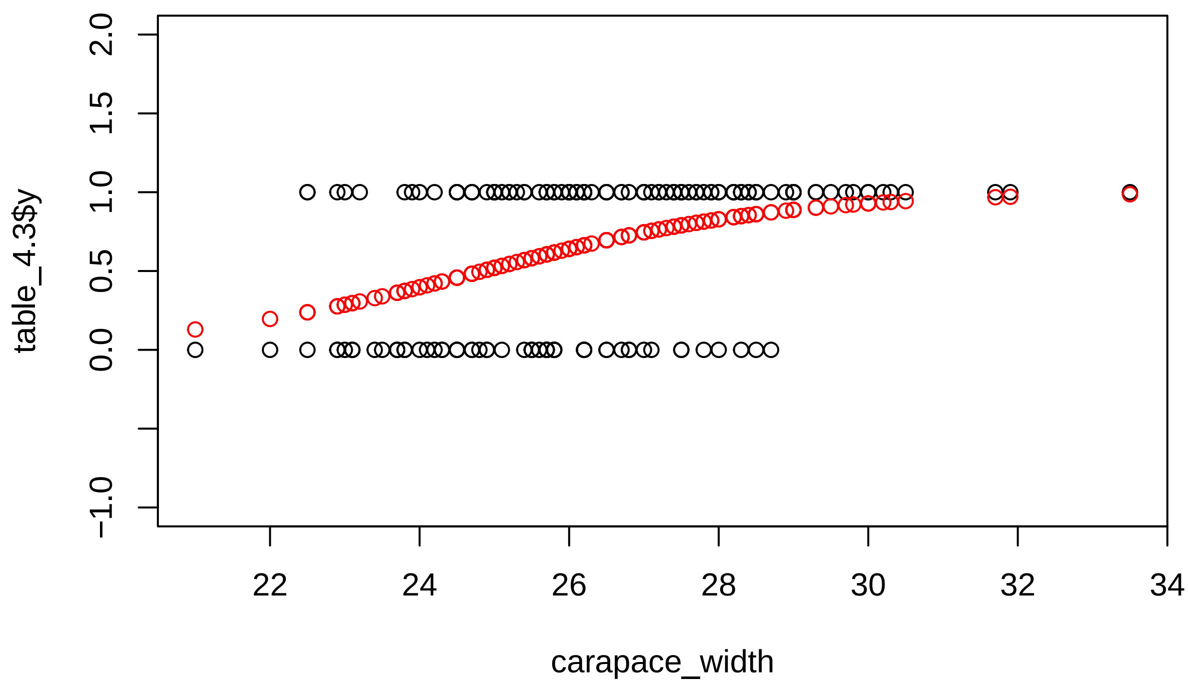
$$\pi(x) = \frac{exp(\alpha + \beta x)}{1 + exp(\alpha + \beta x)}.$$

Here $\pi(x)$ denote the probability that a female horseshoe crab of width x has a satellite. The ML fit is

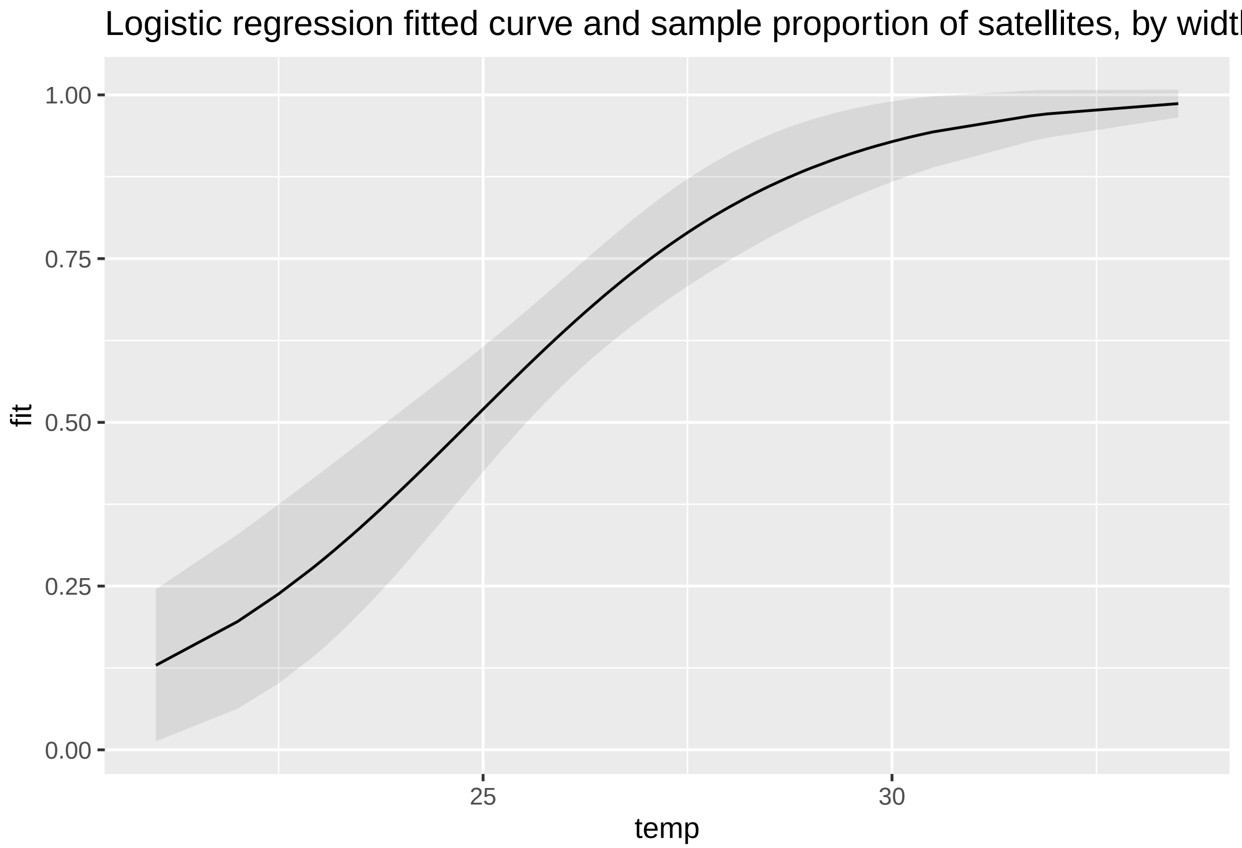
$$\hat{\pi}(x) = \frac{exp(-12.351 + 0.497x)}{1 + exp(-12.351 + 0.497x)}.$$

Substituting $x = 26.3$cm which is the mean width in this sample, $\hat{\pi}(x) = 0.674$. The estimated probability is $\frac{1}{2}$ when $x = \frac{-\hat{\alpha}}{\beta} = \frac{12.351}{0.497} = 24.8$.

```
plot(carapace_width,table_4.3$y,ylim=c(-1,2))
points(carapace_width,logistic.reg.model$fitted.values,col="red")
```




```
library('tibble')
library('ggplot2')
ndata<-data.frame(temp=carapace_width)
## add the fitted values by predicting from the model for the new data
ndata <- add_column(ndata, fit = predict(logistic.reg.model, type = 'response'))
ndata <- add_column(ndata,se = predict(logistic.reg.model,type = 'response',se.fit = TRUE)$se.fit)
ndata <- add_column(ndata, upr = ndata$fit + 2 * ndata$se, lwr = ndata$fit - 2 * ndata$se)
ggplot(ndata, aes(x = temp, y = fit)) +geom_line()+
  geom_ribbon(data = ndata, aes(ymin = lwr, ymax = upr),alpha = 0.1)+labs(title="Logistic regression fi
```



At the mean width, $\hat{\pi}(x) = 0.674$, and $\hat{\pi}(x)$ increases by about $\hat{\beta}[\hat{\pi}(x)(1-\hat{\pi}(x))] = 0.497(0.674)(0.326) = 0.11$ for a 1 cm increase in width. Or, we could report $\hat{\pi}(x)$ at the quartiles of x. The lower quartile, median, and upper quartile for width are 24.9, 26.1, and 27.7; $\hat{\pi}(x)$ at those values are 0.51, 0.65, and 0.81, increasing by 0.30 over the x values for the middle half of the sample.